

LocalLens

Local Guide Marketplace App

"Explore like a local. See what maps cannot show."

Version	1.0.0
Date	April 2026
Type	Full Stack Mobile App
Stack	React + Node.js + PostgreSQL

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1. Executive Summary

LocalLens is a peer-to-peer marketplace connecting travelers with local residents who act as personal city guides, photographers, and experience curators.

The core idea is simple: every city has locals who know it inside out — the hidden cafes, the secret viewpoints, the street food no tourist ever finds. LocalLens turns these locals into guides, giving travelers authentic experiences while giving locals a way to earn from their knowledge.

Metric	Value
Target Market	India — Travelers + Local Guides
Primary Cities	Bengaluru, Mumbai, Jaipur, Goa, Delhi
Revenue Model	10% platform fee per booking
App Type	Mobile-first (iOS + Android via React)
MVP Timeline	2-3 months
Tech Stack	React, Node.js, PostgreSQL, Socket.IO, Razorpay

The Problem We Solve

Most travel apps show you the same tourist spots everyone else visits. Google Maps shows you restaurants — not which ones the locals actually eat at. TripAdvisor shows you reviews — not a human who will walk with you. LocalLens fills this gap by connecting real people.

The Opportunity

- India domestic tourism market: \$150B+ annually
- Growing demand for authentic, personalized travel experiences
- No dominant player in the peer-to-peer local guide space in India
- Gig economy growth — locals want flexible income opportunities
- Post-COVID travelers prioritize unique, off-the-beaten-path experiences

2. App Concept & User Roles

LocalLens has two types of users — Travelers and Guides — each with their own journey through the app.

Two Sides of the Marketplace

Role	Who They Are	What They Do	How They Benefit
Traveler	Tourist or visitor to a city	Browse guides, book tours, explore hidden gems	Authentic local experience at affordable prices
Local Guide	City resident with local knowledge	Create profile, set prices, accept bookings	Earn money from their city knowledge
Photographer Guide	Local with photography skills	Offer photo tours + portrait sessions	Premium pricing for photography services

Core User Journeys

Traveler Journey

- 1. Download & Sign Up:** Register as Traveler, set interests and home city
- 2. Discover Guides:** Browse guides by city, category, price, and rating
- 3. Watch Reels:** See short videos of places guides take people
- 4. Book a Guide:** Select date, duration, and pay securely via Razorpay
- 5. Meet & Explore:** Live tracking shows guide arrival, explore hidden gems
- 6. Review & Share:** Leave review, post your own reel of the experience

Guide Journey

- 1. Register as Guide:** Create profile with bio, city, languages, expertise tags
- 2. Set Pricing:** Set hourly, half-day, and full-day rates
- 3. Go Online:** Toggle availability to start accepting bookings
- 4. Accept Bookings:** Confirm or decline booking requests from travelers
- 5. Conduct Tour:** Meet traveler, share live location, show hidden gems
- 6. Get Paid:** Earnings credited to wallet after tour completion

3. App Screens & Features

LocalLens consists of 11 screens covering the complete user journey for both travelers and guides.

Screen	Key Features
Onboarding / Splash	Role selection (Traveler / Guide), login and registration with JWT auth
Traveler Home Feed	Guide discovery with search, category filters, top guides carousel, group tours banner
Guide Profile	Full profile with cover photo, gallery, reviews, pricing, book/message CTAs
Booking Flow	3-step: date picker, duration selector, price summary, Razorpay payment
Reels Feed	Vertical video feed (Instagram-style) from guides and travelers
Map / Explore	Interactive map with live guide pins, hidden gem locks, group tour markers
Group Tours	Browse and join shared tours, participant avatars, group chat
Messages / Chat	Real-time chat per booking via Socket.IO, typing indicators, read receipts
Guide Dashboard	Earnings stats, availability toggle, upcoming bookings, review management
Guide Registration	Multi-step onboarding: city, bio, expertise tags, pricing, availability
Profile Screen	User stats, settings menu, logout, become-a-guide CTA

Key Feature Deep Dives

Hidden Gems Map

Each guide can pin secret locations on the city map. These pins appear locked to all travelers. When a traveler books a guide, the gems associated with that guide become unlocked and visible. This creates curiosity, drives bookings, and gives guides a powerful differentiator.

Reels & Video Discovery

Guides and travelers post short vertical videos tagged to locations. Guide reels include a "Book This Guide" CTA button. Traveler reels build social proof. The feed algorithm surfaces reels from nearby cities when in "Near Me" mode.

Group Tours

Guides create open tour slots that multiple travelers can join, splitting the cost. This reduces the price per person, increases guide earnings, and helps solo travelers meet other people. Each group tour has its own real-time group chat room.

4. Technology Stack

LocalLens is built on a modern, scalable full-stack architecture designed for mobile-first experiences.

Frontend

Technology	Version	Purpose
React	18.3.1	UI component framework
React Router	7.13.0	Client-side navigation
Tailwind CSS	4.1.12	Utility-first styling
shadcn/ui	Latest	Pre-built UI components
Lucide React	0.487.0	Icon library
Socket.IO Client	4.x	Real-time communication
Vite	6.3.5	Build tool and dev server
TypeScript	5.x	Type safety

Backend

Technology	Version	Purpose
Node.js	20+	Runtime environment
Express.js	4.18	REST API framework
Prisma ORM	5.22	Database access layer
PostgreSQL	15+	Primary database (via Neon/Supabase)
Socket.IO	4.6	Real-time WebSocket server
JWT	9.0	Authentication tokens
Bcrypt	2.4	Password hashing
Zod	3.22	Request validation
Razorpay	2.9	Payment processing (India)
Cloudinary	2.0	Image and video storage
Twilio	4.23	SMS and OTP

5. Database Architecture

LocalLens uses a PostgreSQL database with 16 tables covering all entities in the marketplace.

Database Tables

Table	Description	Key Fields
users	All users (guides + travelers)	id, email, role, firebaseUid
guide_profiles	Guide-specific data	userId, city, pricing, avgRating, isAvailable
traveler_profiles	Traveler-specific data	userId, interests, loyaltyPoints, walletBalance
bookings	All tour bookings	guideId, travelerId, status, totalAmount, escrowReleased
payments	Payment records	bookingId, razorpayOrderId, status, amount
messages	Chat messages per booking	bookingId, senderId, content, isRead
reviews	Post-tour reviews	bookingId, rating, comment, guideResponse
reels	Video posts	userId, videoUrl, city, reelType, likesCount
reel_likes	Reel engagement	reelId, userId
reel_comments	Reel comments	reelId, userId, comment
group_tours	Group tour listings	guideId, maxMembers, pricePerPerson, date
group_tour_members	Tour participants	groupTourId, userId, paymentStatus
hidden_gems	Secret locations by guides	guideId, latitude, longitude, isLocked
notifications	Push notifications	userId, title, type, isRead
wallet_transactions	Earnings and payments	userId, amount, type, description
sos_alerts	Emergency alerts	userId, latitude, longitude, isResolved

6. API Endpoints

The backend exposes a RESTful API running on port 5001 with JWT authentication.

Authentication

Method	Endpoint	Description
POST	/api/auth/register	Create account (traveler or guide)
POST	/api/auth/login	Login, returns JWT tokens
POST	/api/auth/refresh	Refresh expired access token
POST	/api/auth/send-otp	Send OTP via SMS (Twilio)

Guides

Method	Endpoint	Description
GET	/api/guides	Search guides (city, category, price filter)
GET	/api/guides/:id	Guide profile + reviews + reels
POST	/api/guides/register	Create guide profile
PATCH	/api/guides/availability	Toggle online/offline status
PATCH	/api/guides/location	Update live GPS location
POST	/api/guides/hidden-gems	Add hidden gem pin to map
GET	/api/guides/dashboard/stats	Earnings, bookings, ratings

Bookings

Method	Endpoint	Description
POST	/api/bookings	Create new booking
GET	/api/bookings/my	My bookings (as guide or traveler)
PATCH	/api/bookings/:id/status	Confirm or cancel booking
PATCH	/api/bookings/:id/complete	Complete tour + release escrow

Payments

Method	Endpoint	Description
POST	/api/payments/create-order	Create Razorpay order
POST	/api/payments/verify	Verify payment signature
POST	/api/payments/webhook	Razorpay webhook handler

Reels

Method	Endpoint	Description
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GET	/api/reels	Feed (For You / Near Me / Following)
POST	/api/reels	Upload new reel
POST	/api/reels/:id/like	Like or unlike a reel
POST	/api/reels/:id/view	Increment view count

7. Real-Time Features & Payment Flow

LocalLens uses Socket.IO for real-time communication and Razorpay for secure payments.

Socket.IO Events

Event (Emit)	Event (Listen)	Purpose
chat:join	—	Join booking chat room
chat:message	chat:new-message	Send and receive messages
chat:typing	chat:typing	Show typing indicator
chat:mark-read	chat:messages-read	Mark messages as read
location:update	location:guide-update	Guide sends live GPS to traveler
tour:started	tour:status	Tour begins — tracking starts
group-tour:join	—	Join group tour room
group-tour:message	group-tour:new-message	Group tour chat
—	user:X:new-booking	Guide receives new booking alert
—	user:X:payment-confirmed	Guide notified of payment
—	user:X:tour-complete	Traveler prompted to review

Razorpay Payment Flow

- Step 1:** Traveler confirms booking details and clicks "Confirm & Pay"
- Step 2:** Frontend calls POST `/api/payments/create-order` with `bookingId`
- Step 3:** Backend creates Razorpay order, returns `orderId` + amount
- Step 4:** Frontend opens Razorpay checkout modal (UPI, card, netbanking)
- Step 5:** Traveler completes payment on Razorpay
- Step 6:** Frontend calls POST `/api/payments/verify` with signature
- Step 7:** Backend verifies HMAC signature — if valid, booking status = CONFIRMED
- Step 8:** Guide receives real-time notification of confirmed booking
- Step 9:** After tour, guide clicks Complete — escrow released to guide wallet
- Step 10:** Platform keeps 10% fee, guide receives 90% of base price

Security Measures

- JWT access tokens (7 day expiry) + refresh tokens (30 days)
- Bcrypt password hashing with 12 salt rounds
- Razorpay HMAC-SHA256 signature verification on every payment
- Helmet.js HTTP security headers on all API responses

- Rate limiting: 100 requests per 15 minutes per IP
- Escrow system: guide only paid after tour is marked complete
- SOS button sends GPS coordinates + triggers emergency SMS via Twilio

8. Development Roadmap

LocalLens is being built in 3 phases to ensure a fast launch followed by iterative growth.

Phase	Timeline	Features	Status
Phase 1 MVP Launch	Months 1-3	Guide + traveler profiles Search & discovery Booking system Razorpay payments In-app chat Reviews & ratings Basic map with guide pins	In Progress
Phase 2 Growth	Months 3-5	Reels & video posts Group tours Live GPS tracking Hidden gems map AI guide matching Loyalty points system Verified guide badges	Planned
Phase 3 Scale	Months 6+	AI itinerary builder Multilingual chat translation Corporate/agency API Offline mode Background checks City expansion India-wide	Future

Current Progress

Feature	Status
Figma Design (11 screens)	Complete
Backend API (12 route files)	Complete
Database Schema (16 tables)	Complete
Frontend React App	Complete
Frontend-Backend Integration	Complete
Database Migration (Neon)	Complete
Real-time Chat (Socket.IO)	Built, testing
Razorpay Payment UI	Built, needs keys
Google Maps Integration	Planned
Video Upload (Cloudinary)	Planned
App Deployment	Planned

9. Deployment & Infrastructure

LocalLens is designed to be deployed on free/low-cost cloud infrastructure for the MVP phase.

Recommended Hosting Stack

Service	Provider	Cost	Purpose
Backend API	Railway or Render	Free tier	Node.js server
Database	Neon.tech	Free tier	PostgreSQL database
Video/Image Storage	Cloudinary	Free tier (25GB)	Reels + guide photos
Frontend	Vercel or Netlify	Free tier	React app hosting
Real-time	Same as backend	Included	Socket.IO server
Payments	Razorpay	2% per transaction	UPI/card payments
SMS/OTP	Twilio	Pay per use	OTP verification
Push Notifications	Firebase FCM	Free tier	Mobile notifications

Environment Variables Required

Variable	Description
DATABASE_URL	PostgreSQL connection string (Neon/Supabase)
JWT_SECRET	Secret key for access token signing
JWT_REFRESH_SECRET	Secret key for refresh token signing
RAZORPAY_KEY_ID	Razorpay API key (from dashboard)
RAZORPAY_KEY_SECRET	Razorpay secret key
CLOUDINARY_CLOUD_NAME	Cloudinary account name
CLOUDINARY_API_KEY	Cloudinary API key
CLOUDINARY_API_SECRET	Cloudinary API secret
TWILIO_ACCOUNT_SID	Twilio account SID
TWILIO_AUTH_TOKEN	Twilio auth token
FIREBASE_PROJECT_ID	Firebase project ID for FCM
PLATFORM_FEE_PERCENT	Platform commission (default: 10)

Local Development Setup

```
# Clone/unzip backend and frontend
cd locallens-backend
npm install --legacy-peer-deps
cp .env.example .env # Fill in your values
npm run db:generate
```

```
npm run db:migrate
npm run dev # Runs on port 5001

# In second terminal
cd locallens-app
npm install --legacy-peer-deps
npm run dev # Runs on port 5173
```

10. Business Model & Monetization

LocalLens generates revenue through multiple streams designed to scale with usage.

Revenue Streams

Revenue Stream	How It Works	Typical Amount
Platform Commission	10% fee on every booking automatically deducted	10% of booking value
Guide Pro Badges	Guides pay monthly for higher visibility in search	Rs. 299-599/month
Featured Listings	Guides pay to appear at top of search results	Rs. 499-999/month
Group Tour Fees	Same 10% commission on group tour bookings	10% per participant
Corporate API	Travel agencies white-label the platform	Custom enterprise pricing
Photography Premiums	Higher commission on photography sessions	12-15% commission

Unit Economics Example

For a half-day tour booked at Rs. 1,200:

Party	Amount	Notes
Traveler pays	Rs. 1,320	Base price + 10% platform fee
Platform fee (10%)	Rs. 120	LocalLens revenue
Guide receives	Rs. 1,200	Credited to guide wallet after completion
Guide payout	Rs. 1,200	Transferred to bank via UPI/NEFT

Competitive Advantage

- First-mover advantage in India peer-to-peer local guide space
- Reels-based discovery — visual content drives bookings better than text
- Hidden gems feature creates unique FOMO and differentiates from Google Maps
- Group tours reduce cost barrier for travelers, increase earnings for guides
- AI matching removes friction in finding the right guide
- Escrow payments build trust on both sides of the marketplace
- SOS safety feature — critical for solo female traveler segment

LocalLens — Built with passion for authentic travel experiences.

For questions contact: Nigamraj | April 2026